Allison Zhao

allison-stack | Allison Zhao | allisonzhao.uni@gmail.com | 289-775-9755 | allisonzhao.vercel.app

Education

McMaster University Sept. 2022 – May 2027
Bachelor of Applied Science in Computer Science with Co-op ***3.9/4.0 GPA***

- **Achievements:** Deans' Honour List, The Dr. Harry Lyman Hooker Scholarship, The Pollock Family Academic Grant, The McMaster University Award of Excellence
- **Extracurriculars:** DeltaHacks VP Logistics, McMaster Start Coding Mentor

Skills

Languages: Python, SQL, C/C++, TypeScript/JavaScript, Java, HTML, CSS, Haskell, Elm
Frameworks and Libraries: NextJS, React, Tailwind CSS, LangGraph, FastAPI, Vite
Tools: Git/GitHub, Linux, Bash Scripting, PostgreSQL, MySQL, Google Cloud Platform, Docker

Experience

DevOps Engineer Intern May 2025 – Present
Huawei Technologies Canada

- Automated TPC-H power/throughput benchmark execution for GaussDB, a cloud-native distributed relational database, by developing Bash scripts to compile per-query QphH scores and timing data; contributed to a 30TB **world record** of 39.974M QphH (+43% over prior record) and 10TB record of 16.2M QphH using half the compute of the prior record holder
- Identified **6 pre-release bugs** in TaurusDB Serverless's online-resizable buffer pool by designing a functional verification test plan covering concurrent resize, in-progress checkpoints, dirty pages, vacuum, and master/standby replication configs
- Restored regression test coverage for TaurusDB for PostgreSQL's Storage Abstraction Layer by debugging and remediating **30+** failing test cases — fixing broken logic, porting MySQL-specific tests to PostgreSQL compatibility, and resolving codebase drift since 2018; tests integrated into the development team's CI pipeline

Undergraduate Research Assistant May 2024 – Aug. 2024
McMaster University

- Implemented **168 RISC-V instructions** across 4 ISA extensions (RV32/64I, RV32/64M, RV32F, RV64D) in Haskell by translating official ISA specifications into formal type-class semantic models covering both general-purpose and floating-point register files
- Achieved **100% ISA compliance** by building a Dockerized QEMU test harness of 82 C programs that cross-validated instruction output against the Haskell interpreter, resolving correctness bugs in signed/unsigned overflow, division-by-zero, and floating-point bit-exact transfer edge cases
- Implemented 3 vectorized **IBM MASS** math library functions (logarithm, power, cube root) targeting POWER ISA SIMD vector instructions, applying Horner's method polynomial approximation and fused multiply-add error compensation to achieve IEEE 754-accurate single-precision results

Projects

leet-buddy | *TypeScript, PostgreSQL (Supabase), Vite, Chrome APIs*

- Built a Chrome extension that locks LeetCode hints behind difficulty-scaled timers and enforces approach-first prompts before coding
- Implemented SM2 spaced-repetition for problem review scheduling and a real-time challenger mode with friend racing backed by Supabase

ForgeLab | *Python, FastAPI, LangGraph, React, Docker*

- Designed a 7-agent SE pipeline using LangGraph where a routing agent dynamically assembles per-task workflows (research → plan → code → review → verify) instead of running every stage
- Local-first by default (Ollama); agents produce diffs only, never modify the codebase directly; test execution sandboxed in Docker